

Andrey Sarantsev

University of Nevada, Reno

Department of Mathematics & Statistics

Davidson Mathematics & Science Center 234

Work Email: asarantsev@unr.edu

Department Phone: (775) 784-6773

asarantsev.github.io/WebArchive/

RESEARCH INTERESTS

Statistics, Probability, Quantitative Finance

- Point estimation for statistical distributions
- Financial econometrics
- Astronomy data analysis
- Forest dynamics
- Systemic financial risk
- Risk theory and ruin probability
- Random particle systems interacting through ranks
- Long-term stability of stochastic processes
- Concentration of measure for stochastic equations
- Stochastic portfolio theory

EMPLOYMENT

University of Nevada, Reno (UNR)

Department of Mathematics and Statistics

Associate Professor (with tenure), Jul 2024 – now

Assistant Professor (tenure-track), Jul 2018 – Jun 2024

University of California, Santa Barbara (UCSB)

Department of Statistics and Applied Probability

Visiting Assistant Professor, Sep 2015 – Jun 2018

Mentor: JEAN-PIERRE FOUQUE. Partially supported by his NSF grant DMS 1409434

EDUCATION

University of Washington, Seattle (UW)

Ph.D. in Mathematics, Sep 2010 – Jun 2015

Adviser: SOUMIK PAL. **Thesis:** Competing Brownian Particles

Lomonosov Moscow State University, Moscow, Russia

Specialist (Master's equivalent) with Honors in Mathematics, Sep 2005 – Jun 2010

Undergraduate Mentor: VLADIMIR PITERBARG

57th mathematics high school

Top math high school in Moscow, Russia, 2001–2005

RESEARCH ADVISING

Ph.D. students: Smaila Amoanu, Francis Boampong, Cornelius Benle (current), Abraham Atsiwo (Statistics, Aug 2024), Jihyun Park (Statistics, Aug 2024), Hayden Brown (Applied Math, May 2024)

M.S. students: Kwame Boamah-Addo (Statistics, May 2022), Hayden Brown (Applied Math, May 2021)

Undergraduate students: 13 former students: Brandon Flores, Jaucelyn Canfield, Taran Grove, Akram Reshad, Lissa Callahan, Melissa Eid, Chyna Metz-Bannister, Franklin Fuchs, Michael Reyes, Peter Kilonzo, Mohagoney Moore, Angel Piotrowski, Ian Anderson

SUBMITTED MANUSCRIPTS

1. Fast Convergence for Weighted Least Squares Estimates (2026). arXiv:2605.00198.
2. Random Stability of Random Variables (2026). arXiv:2603.28093.
3. Modeling Stock Returns and Volatility Using Bivariate Gamma Generalized Laplace Law (2026). With TOMASZ KOZUBOWSKI, JAMES SPIKER. arXiv:2605.00196.
4. GPS Constellation Search for Exotic Physics Messengers Coincident with the Binary Neutron Star Merger GW170817 (2026). With ARKO SEN, GEOFFREY BLEWITT, PAUL RIES, ANDREI DEREVIANKO. arXiv:2602.15296.
5. Valuation Measure of the Stock Market using Stochastic Volatility and Stock Earnings (2026). With ANGEL PIOTROWSKI and IAN ANDERSON. arXiv:2508.06010.
6. Capital Asset Pricing Model with Size Factor and Normalizing by Volatility Index (2026). With ABRAHAM ATSIWO. arXiv:2410.22498.
7. VIX as Stochastic Volatility for Corporate Bonds (2026). With JIHYUN PARK. arXiv:2410.22498.
8. Log Heston Model for Monthly Average VIX (2025). With JIHYUN PARK. arXiv:2410.22471.

PUBLISHED ARTICLES

1. Tutorial on Running Median Subtraction Filter with Application to Searches for Exotic Field Transients in Multi-Messenger Astronomy (2025). With GEOFFREY BLEWITT, ANDREI DEREVIANKO, ARKO SEN. *AIP Advances* **15** 055205. arXiv:2410.03773.
2. A New Stock Market Valuation Measure with Applications to Retirement Planning (2025). *Statistics and Risk Modeling* **42** (1–2), 1–18. arXiv:1905.04603
3. The Variance-Gamma Distribution: A Review (2025). With ADRIAN FISCHER and ROBERT E. GAUNT. *Statistical Science* **40** (2), 235–258. arXiv:2303.05615.
4. Boundary Approximation for Sticky Jump-Reflected Processes on the Half-Line (2024). With ANDREY PILIPENKO. *Electronic Journal of Probability* **29**, 1–21. arXiv:2303.02771.
5. Modified Method of Moments for Generalized Laplace Distributions (2024). With ADRIAN FISCHER and ROBERT E. GAUNT. *Communications in Statistics: Simulation and Computation* 1–18. arXiv:2203.10775.
6. IID Time Series Testing (2023). *Theory of Stochastic Processes* **27** (1) 41–52. arXiv:2203.10405.
7. Birth and Death Processes in Interactive Random Environments (2022). With GUODONG PANG and YURI SUHOV. *Queueing Systems* **102** (1–2), 269–307. arXiv:2203.10411.
8. Transient Behaviors of Single-Server Queues with Diffusive Rates (2022). With GUODONG PANG and YURI SUHOV. *Queueing Systems* **100** (3–4), 333–335.
9. Penalty Method for Obliquely Reflected Diffusions (2021). *Lithuanian Mathematics Journal* **61** (4), 518–549. arXiv:1509.01777.
10. Optimal Portfolio with Power Utility for Absolute and Relative Wealth (2021). *Stat. & Probab. Let.* **179** 109225. arXiv:2105.0813.
11. A Stock Market Model Based on CAPM and Market Size (2021). With BRANDON FLORES and BLESSING OFORI-ATTA. *Annals of Finance* **17** (3), 405–424. arXiv:1907.08911.
12. Sub-exponential Rate of Convergence to Equilibrium for Processes on the Half-line (2021). *Statistics and Probability Letters* **175** 109115. arXiv:2003.10614.
13. Time Series Analysis of Forest Dynamics at the Ecoregion Level (2020). With OLGA RUMYANTSEVA and NIKOLAY STRIGUL. *Forecasting* **2** (3), 364–386.

14. Convergence Rate to Equilibrium in Wasserstein Distance for Reflected Jump–Diffusions (2020). *Statistics and Probability Letters* **165** 108860. arXiv:2003.10590.
15. Stationary Distributions and Convergence for M/M/1 Queues in Interactive Random Environment (2020). With YANA BELOPOLSKAYA, GUODONG PANG, and YURI SUHOV. *Queueing Systems* **94** (3–4), 357–392. arXiv:1902.03941.
16. A Note on Jump Atlas Models (2020). With CLAYTON BARNES. *Brazilian Journal of Probability and Statistics* **34** (4), 844–857. arXiv:1610.04323.
17. Autoregression Modeling of Forest Dynamics (2019). With OLGA RUMYANTSEVA and NIKOLAY STRIGUL. *MDPI Forests* **10** (12), 1074. arXiv:1911.09182.
18. Exponential Convergence Rate of Ruin Probabilities for Level-Dependent Lévy-Driven Risk Processes (2019). With PIERRE-OLIVIER GOFFARD. *Journal of Applied Probability* **56** (4), 1244–1268. arXiv:1710.01845.
19. Talagrand Concentration Inequalities for Stochastic Partial Differential Equations (2019). With DAVAR KHOSHNEVISAN. *Stochastic Partial Differential Equations: Analysis and Computations* **7** (4), 679–698. arXiv:1709.07098.
20. Stationary Distributions and Convergence of Walsh Diffusions (2018). With TOMOYUKI ICHIBA. *Bernoulli* **25** (4A), 2439–2478. arXiv:1706.07127.
21. Dynamic Contagion in a Banking System with Births and Defaults (2019). With TOMOYUKI ICHIBA and MICHAEL LUDKOVSKI. *Annals of Finance* **15** (4), 489–538. arXiv:1807.08987.
22. Comparison Techniques for Competing Brownian Particles (2019). *Journal of Theoretical Probability* **32** (2), 545–585. Available at arXiv:1305.1653.
23. Brownian Particles with Rank-Dependent Drifts: Out-of-Equilibrium Behavior (2019). With MANUEL CABEZAS, AMIR DEMBO, VLADAS SIDORAVICIUS. *Communications in Pure and Applied Mathematics* **72** (7), 1424–1458. arXiv:1708.01918.
24. Large Rank-Based Models with Common Noise (2019). With PRAVEEN KOLLI. *Stat. Probab. Let.* **151**, 29–35. arXiv:1802.06202
25. A Note on Transportation Cost Inequalities for Diffusions with Reflections (2019). With SOUMIK PAL. *Electronic Communications in Probability* **24** (21), 1–11. arXiv:1808.02164.
26. Modeling Systemic Risk with Interbank Flows, Borrowing, and Investing (2018). With ADITYA MAHESHWARI. *Risks* **6** (4), 1–26. arXiv:1707.03542.
27. Weak Convergence of Obliquely Reflected Diffusions (2018). *Annals of Institute Henri Poincaré Probability and Statistics* **54** (3), 1408–1431. arXiv:1509.01778.
28. Multiple Collisions in Systems of Competing Brownian Particles (2018). With CAMERON BRUGGEMAN. *Bernoulli* **24** (1), 156–201. arXiv:1309.2621.
29. Infinite Systems of Competing Brownian Particles (2017). *Annals of Institute Henri Poincaré Probability and Statistics* **53** (4), 2279–2315. arXiv:1403.4229.
30. Yet Another Condition for Absence of Collisions for Competing Brownian Particles (2017). With TOMOYUKI ICHIBA. *Electronic Communications in Probability* **22** (8), 1–7. arXiv:1608.07220.
31. Stationary Gap Distributions for Infinite Systems of Competing Brownian Particles (2017). With LI-CHENG TSAI. *Electronic Journal of Probability* **22** (56), 1–20. arXiv:1608.00628.
32. Reflected Brownian Motion in a Convex Polyhedral Cone: Tail Estimates for the Stationary Distribution (2017). *Journal of Theoretical Probability* **30** (3), 1200–1223. arXiv:1509.01781.
33. Two-Sided Infinite Systems of Competing Brownian Particles (2017). *ESAIM Probability and Statistics* **21**, 317–349. arXiv:1509.01859.
34. Explicit Rates of Exponential Convergence for Reflected Jump-Diffusions on the Half-Line (2016). *ALEA Latin American Journal of Probability and Mathematical Statistics* **13** (2), 1069–1093. arXiv:1509.01783.

35. Penalty Method for Reflected Diffusions on the Half-Line (2016).
With CAMERON BRUGGEMAN. *Stochastics* **89** (2), 485-509. arXiv:1509.01776.
36. Diverse Market Models of Competing Brownian Particles with Splits and Mergers (2016).
With IOANNIS KARATZAS. *Annals of Applied Probability* **26** (3), 1329-1361. arXiv:1404.0748.
37. Triple and Simultaneous Collisions of Competing Brownian Particles (2015).
Electronic Journal of Probability **20** (29), 1-28. arXiv:1401.6255.
38. On a Class of Diverse Market Models (2014).
Annals of Finance **10** (2), 291-314. arXiv:1301.5941.

RESEARCH TALKS

- | | |
|------|---|
| 2026 | Joint Mathematics Meetings |
| 2025 | Colorado State University |
| 2024 | Columbia University |
| 2023 | Society for Industrial & Applied Mathematics Conference in Financial Mathematics, Philadelphia |
| 2022 | University of Texas, Dallas; University of Utah |
| 2021 | Frontier Probability Days (University of Nevada Las Vegas) |
| 2020 | University of Montana; Joint Mathematics Meeting; University of Mississippi; Washington State University; Penn State University; European Seminar in Computing; Computational & Methodological Statistics |
| 2019 | American Statistical Association Nevada Sectional Meeting; INFORMS Annual Meeting |
| 2018 | Joint Mathematics Meetings; Florida State University; Cornell University; Carnegie Mellon University; California State University, Los Angeles; University of Nevada, Reno; Frontier Probability Days; University of Minnesota; UCSB; University of Washington; AMS Western and Eastern Fall Sectional Meetings |
| 2017 | AMS Western, Southwestern, and Central Fall Sectional Meetings; INFORMS Annual Meeting in Houston; Center for Financial Mathematics & Actuarial Research (UCSB) 10th anniversary conference; University of Utah; UCSB; Boston University; 9th Western Conference in Mathematical Finance; Seminar on Stochastic Processes; University of Maryland, College Park; University of Delaware; AMS Central Spring Sectional Meeting; University of Washington |
| 2016 | SIAM Conference in Financial Mathematics; Michigan State University; Carnegie Mellon University; Oregon State University; University of Washington; University of Illinois, Chicago; Princeton University; Columbia University; City University of New York |
| 2015 | Southern California Probability Symposium; University of Southern California; UCSB |
| 2014 | Columbia University; Seminar on Stochastic Processes; UCSB |

TEACHING EXPERIENCE

- | | |
|----------------------------|--|
| UNR Undergraduate Classes | Ordinary Differential Equations, Partial Differential Equations, Statistics & Probability for Engineers (large class with TA sections), Probability, Stochastic Processes, Bayesian Inference, Linear Regression |
| UNR Graduate Classes | Probability (Qualifying Exam Class), Stochastic Simulation & Models, Time Series, Quantitative Finance |
| UCSB Undergraduate Classes | Probability, Stochastic Processes |
| UW Undergraduate Classes | Multivariable and Vector Calculus (Calculus III and IV), Differential Equations, Matrix Algebra, Linear Analysis (PDE, systems of ODE), Probability |
| UW Teaching Assistant | Multivariable Calculus (III), REU Inverse Problems |
| UW Homework Grader | Real Analysis (Ph.D. level) |

SERVICE

- Nevada Chapter President of the American Statistical Association (2023)
- Accessibility of mathematical content for visually impaired students
- Organized Department of Mathematics & Statistics Research Library
- Statistics & Data Science graduate committee
- Hiring committee for postdocs
- Graduate committee member for graduate students
- Recommendation letters for graduate applications
- Python workshops for students
- Refereeing research manuscripts
- Portfolio Simulator `asarantsev.pythonanywhere.com`

LANGUAGES AND SOFTWARE

Languages: English (fluent), Russian (native)

Coding: C, Python, R

Editors: $\text{\LaTeX}$, HTML

PERSONAL INFORMATION

Born October 9, 1989, in Moscow, Russia

Citizenship: Russian

USA Permanent Resident (Green Card)

Updated June 16, 2026